

Hengyi Li

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EDUCATION

University of Amsterdam

MSc Mathematics, Stochastics track

Sep. 2023 – Aug. 2026

- **Selected Coursework:** Measure-Theoretic Probability, Stochastic Processes, Stochastic Simulation, Asymptotic Statistics, Forensic Probability and Statistics, Statistical Mechanics, Mathematical Biology, and Data-Driven Decision Making in Operations Research.

University of Liverpool

BSc Applied Mathematics

Sep. 2019 – Jul. 2023

- **Selected Coursework:** Complex Functions, Metric Spaces, Functional Analysis, Ordinary Differential Equations and Control, Partial Differential Equations, Numerical Analysis, Advanced Linear Algebra, and Optimisation Theory.

RESEARCH

Accelerating Convergence to Equilibrium in Gaussian Diffusions

Master's thesis, supervised by Dr. S. G. Cox

Mar. 2026 – Aug. 2026

[\[PDF\]](#) [\[UvA Repository\]](#) [Weighted 7.65/10]

- Studied how noise allocation and non-reversible drift affect convergence of Ornstein–Uhlenbeck diffusions to a prescribed Gaussian equilibrium, reducing Wasserstein convergence to a spectral optimisation problem.
- Gave complete proofs of the two-dimensional optimal-rate results, made the optimal reversible comparison explicit, and determined equality cases for the fixed-drift problem.
- Reformulated the arbitrary-dimensional optimum under $\text{tr}(\mathbf{D}) \leq \eta$ as $\frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1})$ for a general noise budget and constructed an explicit three-dimensional optimiser.
- Compared overdamped, kinetic, and three-level Langevin dynamics and extended synchronous-coupling estimates to uniformly strongly convex non-Gaussian potentials.

Random Walk and the Boundary Value Problems

Bachelor's thesis, supervised by Dr. Y. Liu

Sep. 2022 – May. 2023

[\[PDF\]](#) [Weighted 7.32/10]

- Studied simple random walks on $\mathbb{Z}^d$, including their probability distributions and the recurrence–transience dichotomy across dimensions.
- Used stopping times and exit distributions to represent discrete Dirichlet solutions and prove uniqueness on finite domains and for bounded solutions under an exit-time condition.

SELECTED ACADEMIC PROJECTS

Superinfections Can Induce Evolutionarily Stable Coexistence of Pathogens

Course essay, Mathematical Biology

Nov. 2025 – Jan. 2026

[\[PDF\]](#)

- Derived invasion exponents and coexistence conditions in a nested within- and between-host superinfection model, illustrating evolutionary branching towards stable dimorphism.

Lee–Yang Theorem: Foundations, Proof and Applications

Course essay, Statistical Mechanics, supervised by Prof. dr. W. M. Ruszel Dec. 2024 – Jan. 2025
[\[PDF\]](#)

- Developed a self-contained complex-analytic exposition showing how zero-free finite-volume partition functions yield analytic continuation of the pressure.
- Explained the circle theorem and the analyticity of connected correlations.

Evaluating String Theory Through the Lens of Scientific Realism

Course essay, Philosophy of Physics, co-authored with O. Shah Nov. 2024 – Dec. 2024
[\[PDF\]](#)

- Compared the no-miracles, no-alternatives, and unexpected-explanatory-coherence arguments with constructive empiricism and pessimistic meta-induction, arguing that string theory aligns most naturally with structural realism under current empirical limitations.

China Undergraduate Mathematical Contest in Modeling

Team modelling project Sep. 2019

- Developed differential-equation and optimisation models for pressure regulation in a high-pressure fuel-pipe system and simulated control strategies for the pump and pressure-reducing valve.

EXPERIENCE

Ping An Insurance Company of China

Intern, Motor Insurance Department Jul. 2022

- Cleaned and summarised historical motor-insurance data; computed variance, covariance, empirical distributions, and 95% Value at Risk (VaR).
- Explored regression models for motor-insurance risk factors and used Monte Carlo simulations to compare model outputs with historical data.

Daqing High-Tech City Construction Investment Co., Ltd.

Intern, Construction and Economics Department Jun. 2022

- Prepared labelled spatial-feature data and tested SVM-based classification workflows for a smart-city project.

Kaiyuan Securities

Intern, Securities Investment Advisory Jul. 2020 – Aug. 2020

- Used Python to summarise historical data for major asset classes in a simulated investment-analysis exercise.

SKILLS & INTERESTS

Tools	LaTeX, Python, Git, Linux/Bash, MATLAB, Java, Hugin.
Languages	Chinese (native), English (language of instruction), Japanese (fluent).
Beyond Mathematics	Literature, philosophy, history, painting, cinema, and experimental video.